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Overcoming Challenges of Drilling Through a Sour Overpressured Shallow Aquifer in a Sabkha Environment: A Case Study from North-Central Oman

Shahab Al Amri, Nadir Al Rawahi, Ibrahim Al Rahbi, and Nafa Al Jahdhami, Shell Development Oman, Muscat, Oman

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Abstract

Drilling through shallow overpressured aquifers in sabkha environments presents a unique set of technical and safety challenges. These include high pore pressure gradients, unstable formations, and the presence of sour gas (H_2S), all of which can significantly increase operational risk and cost. This paper presents a case study from Shell Development Oman's operations in the Umm as Samim Sabkha in North-Central Oman, where multiple wells encountered artesian flow and H_2S hazards during surface drilling. A detailed investigation revealed that traditional planning underestimated pore pressures and failed to account for localized aquifer behavior.

By integrating geological, operational, and hydrogeological data, the team developed an updated pore pressure prediction model and implemented targeted mitigation strategies, including optimized cementing practices and contingency planning. These measures led to the successful drilling of a subsequent well without incident, reducing non-productive time by 67%. The study demonstrates the value of a data-driven, cross-disciplinary approach in managing complex shallow hazards and provides a framework for safer and more efficient drilling in similar environments.

Introduction

Shell Development Oman (SDO) currently operates deep tight gas-condensate reservoirs in Blocks 10 & 11 within North-Central Oman. This paper focuses on a specific area where the SDO is currently drilling exploration wells located within the Umm as Samim Sabkha region (Fig. 1). The Umm as Samim, located in central Oman, is an expansive salt flat where several wadis from the northern mountains converge. Due to its extreme salinity, the region supports very little vegetation (König, 2012).

Drilling in this environment is further complicated by the presence of shallow, overpressured aquifers that are likely to be sour. These aquifers exhibit high pressure gradients and unpredictable fluid properties, increasing the risk of well control incidents. Historically, the region has had limited subsurface data and few control points, making it difficult to anticipate and mitigate drilling hazards. Recent campaigns have encountered severe artesian flow incidents while drilling through the shallow section, stressing the need for a more robust and informed approach.



Figure 1—Satellite map of Umm as Samim Sabkha area within SDO's operated Blocks 10 & 11

This paper presents a case study from SDO's operations in the Umm as Samim Sabkha, where an integrated, data-driven strategy was employed to safely drill through a sour, overpressured shallow aquifer. By combining regional aquifer pressure and fluid property analysis with real-time monitoring and customized well control techniques, the well delivery team successfully mitigated risks, enhanced safety, and improved drilling performance. The insights gained offer valuable guidance for engineers operating in similarly challenging environments.

Geological and Operational Context

Geological Background

The Umm as Samim Sabkha region in North-Central Oman presents a uniquely challenging geological setting for drilling operations. Sabkha soils, commonly found in arid desert regions, are characterized by salt crusts that pose significant challenges for construction and development (Alalawi, 2020). Illustrated in Fig. 2 below, Sabkhas are characterized as low-relief salt flats typically found in arid regions, formed through the accumulation of evaporite minerals (König, 2012, Mountney, 2005). The Umm as Samim, the largest sabkha systems in Oman, is characterized by extreme salinity, minimal vegetation, and unstable soil conditions (Alalawi et al., 2020, König, 2012). These factors complicate surface construction and drilling operations, requiring specialized planning and solutions (Alalawi et al., 2020).

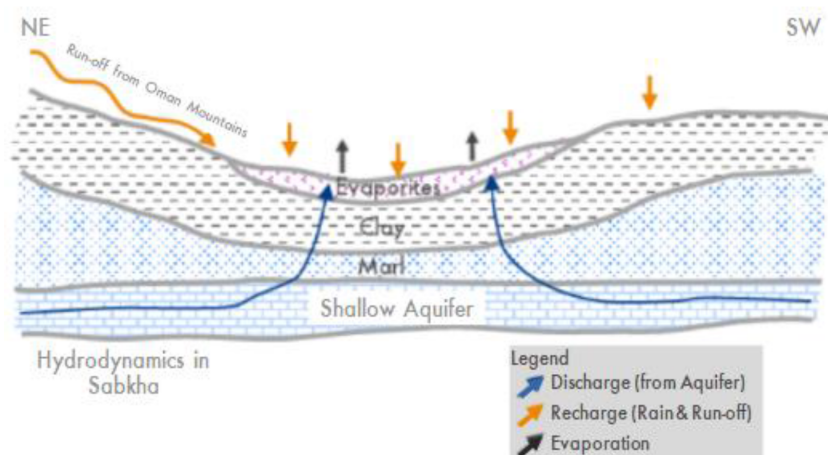


Figure 2—Schematic illustrating the process cycle of evaporite deposits creating the Sabkha



Figure 3—Construction of a well pad for a well within the Sabkha area



Figure 4—Rig substructure stuck in Sabkha location during rig move

Beneath the sabkha surface lie several shallow formations, notably the Rus Formation and the Upper, Middle, and Lower (Shammar) units of the Umm Er Radhuma (UER). These formations act as aquifers and are known for their variable lithology, high porosity zones, and potential sour fluid content. The Rus Formation is primarily composed of evaporites and carbonates and is known to be associated with continuous fluid losses while drilling. The UER Formation, comprised of hard carbonates, is widespread and relatively uniform across Interior Oman. Its permeable and often vuggy middle zones make it a key aquifer in the region (Forbes et al. 2010). While more stable, still pose significant risks due to losses from shallower units, abnormal reservoir pressures and potential for hydrogen sulfide (H_2S) release. The UeR Aquifer is bounded to the North by the Oman Mountains, to the South by the Dhofar Mountains, and to the East by the low lying Huqf outcrops (Al Lamki et al., 1996). It restricted to depths of 250-500 m from which most of the water used for the Oil & Gas industry is extracted. The aquifer has a high likelihood of being sour within the industry likely due to its shallow depth and presence of Sulfate Reducing Bacteria (SRB) which induces high levels of hydrogen sulfide (H_2S). Furthermore, Artesian Flow has also been recorded from wells drilled in the UeR.

Historically, drilling through these shallow formations has been operationally challenging. Frequent incidents of wellbore instability, severe fluid losses, and artesian flows have been reported, often accompanied by H_2S . These challenges are exacerbated by the limited availability of regional subsurface data and the scarcity of offset wells, making planning and preparation difficult. As a result, traditional well planning and drilling approaches have proven inadequate, necessitating a more integrated and predictive strategy to ensure safe and efficient well delivery in this complex environment.

Operational Challenges Overview

Historically, during the well pad construction, 30" conductor pipe is installed 9-meters below ground level. The surface hole is drilled with a 23" bit down to the Arada Formation. An 18-5/8" casing is run and cemented to surface. Surface drilling operations typically encounter two primary risks: drill fluid losses and/or artesian flow. To assess artesian risk, the subsurface team correlated elevation data with observed conditions in offset wells and mapped areas assigning likelihood of artesian flow from shallow aquifer zones as in Fig. 5 below. Their findings indicated that locations with Ground-level Elevation (GLE) less than 61 meters above mean sea level are more susceptible to artesian flow. The target areas to drill the exploration leads, at 60 meters elevation, were flagged as high-risk during planning.

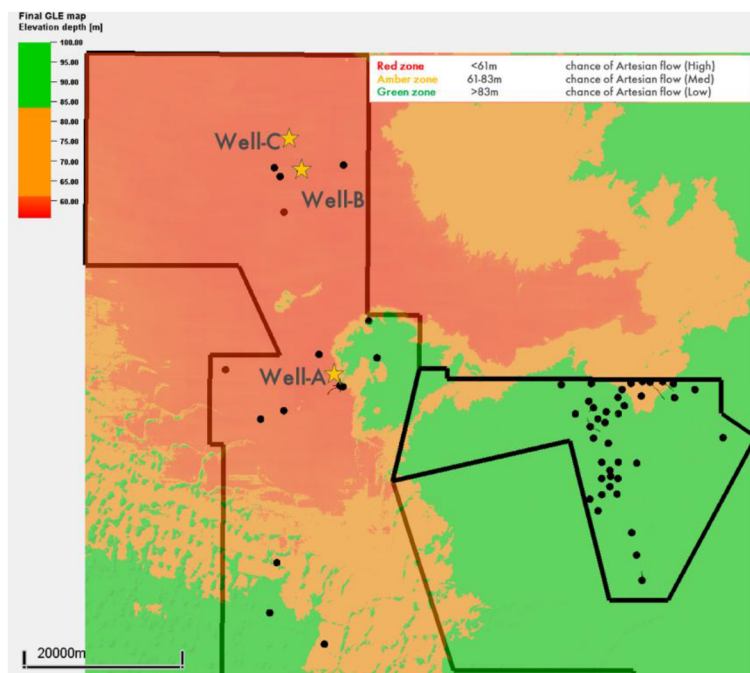


Figure 5—Ground Level Elevation (GLE) Map used to highlight areas potential artesian flow due to proximity from the surface water-table

However, even with precautionary measures implemented ahead of drilling the wells, the first two wells of the exploration campaign suffered from severe artesian flow incidents. The sequence of events and incident summary for each well is described below.

Well-A

Incident Summary. The Well-A pad was constructed at the edge of the Sabkha area (refer to Fig. 5). After drilling the 23" hole and switching returns to the riser at 80 m depth, dynamic losses of 4 m³/hr were observed at 546 m. Static losses of 1 m³/hr were noted during flow check before pulling out of hole (POOH), followed by 2.4 m³/hr while monitoring via trip tank. Upon draining and nipple-down of the 30" riser, resulting in a 104 kPa (15 psi) pressure drop, losses reduced to 0.2 m³/hr, potentially indicating a gain.

While running in hole (RIH) with 18-5/8" casing, losses remained consistent at 0.2 m³/hr. High equivalent circulating densities (ECDs) and surge pressures were suspected during circulation at shoe depth, as dynamic losses resumed at 4 m³/hr.

Cementing operations followed utilizing a Spacer of 10 kPa/m, lead slurry of 13.8 kPa/m while recording with 2.5 m³/hr losses. The cement was displaced with continuous lead slurry observing ~5 m³ of neat cement returns at surface, however, interpretation of the events remained unclear between curing and gain. Upon completing tail displacement and float check operations, the well exhibited sudden artesian flow at 60 m³/hr indicating the well was underbalanced. Flow continued even after pumping 30 m³ of 16.68 kPa/m mud behind casing without success followed by another 60 m³ through PVC pipes behind casing. The cellar pump was continuously operated to redirect flow to a waste pit, followed by discharge to a master pit. Compounding the challenge was a hazardous H₂S concentration (up to 200 ppm), necessitating crew access with breathing apparatus.

Remedial Actions. Following the onset of uncontrolled artesian flow, initial remedial efforts focused on top-fill cementing using fast-setting cement. This attempt failed due to cement washout, attributed to the limited PVC pipe flow rate (0.2 m³/min) being significantly lower than the artesian influx (1 m³/min). PVC deployment was restricted to 30 m on both sides due to pipe stiffness and well geometry.

A subsequent kill operation involved pumping 60 m³ of 16.7 kPa/m fluid, which temporarily reduced flow to 25 m³/hr. However, artesian flow rebounded to 60 m³/hr within 1.5 hours. To further mitigate flow, the landing base clamp was removed, and 6 tons of shot-blasting grit were introduced, reducing flow to 40 m³/hr.

The rig team then modified the 30" riser to enable bullheading via direct access to the casing head housing. During bullheading, 140 m³ of 16.2 kPa/m kill fluid was pumped at 0.75 m³/min, resulting in a diluted mud weight of 12.9 kPa/m. A second attempt involved 35 m³ of 17.95 kPa/m mud, but the observed weight dropped to 10.6 kPa/m, prompting abandonment. Artesian flow was reduced to 17.5 m³/hr.

Final control was achieved through 'spaghetti' pipe deployment. After removing the modified riser, 1.66" pipes were run to 183 m and circulated with 19.2 kPa/m kill fluid, yielding a final mud weight of 18.7 kPa/m. The pipe was retrieved while maintaining annular pressure with heavy mud. Reinstallation of the landing base clamp and PVC pipes enabled a successful top-fill operation, fully controlling the artesian flow. An alternative approach involving temperature logging and squeeze cementing was considered but rejected due to concerns over well integrity and long-term performance. The well remained uncontrolled for ~7.5 days, contributing to significant non-productive time (NPT).

Post-Incident Analysis. Following the artesian flow event, a pressure gauge was installed on the modified riser to capture real-time data from the side entry sub at cellar level. The measured pressure of 214 kPa (31 psi) at surface allowed engineers to back-calculate formation pressure, which was significantly higher than predicted across all horizons. The source of artesian flow could not be definitively identified, but three candidate formations were evaluated in Table 1. All calculated values exceeded the maximum expected pore pressure. Based on knowledge of the regional geology, the flow was assumed to originate from UER Middle given its permeable nature as an aquifer source.

Table 1—List of aquifers suspected to be the source of artesian flow highlighting depth and expected pore pressure based on the surface pressure measurement

Formation	Depth Below Ground-Level	Estimated Pore Pressure (kPa/m)
Rus	74 m	13.6
UER Upper	230 m	11.9
UER Middle	372 m	11.5

In hindsight, the well was clearly underbalanced during all phases of surface drilling relative to the calculated pore pressures, including UER Middle at 11.5 kPa/m as highlighted in Fig. 6 below. The drilling fluid parameters were as follows:

- Equivalent Static Density (ESD): 11.2 kPa/m
- Equivalent Circulating Density (ECD): 11.4 kPa/m
- ESD without riser margin: 11.1 kPa/m

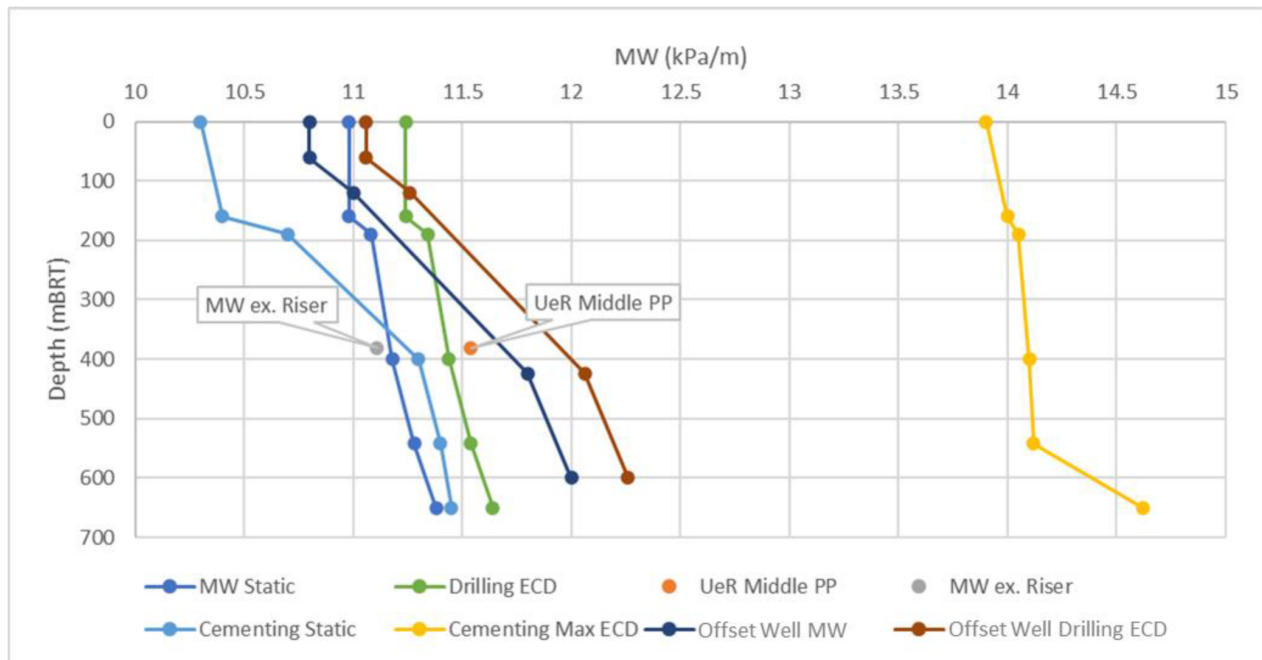


Figure 6—Mud Weight vs. Depth Plot highlighting estimated pore pressure of the UeR Middle compared to drilling parameters

Additionally, draining the 9.14 m riser introduced a pressure drop of 104 kPa (15 psi), equivalent to 0.27 kPa/m at UER Middle, which further exaggerated the underbalance condition prior to cementing. Yet, no artesian flow was observed during drilling or static connections. The onset occurred only after completing the 18-5/8" casing cementing job, suggesting the triggering mechanism was specific to the cementing operation.

The lack of flow earlier in the operation is inconclusive. Possible factors that may have masked or delayed the artesian behavior include:

- **Mud flipping:** Continuous dilution and contamination from artesian water might have altered fluid characteristics, limiting pressure transmission
- **Hydraulic cap effect:** Unconsolidated surface formations or formation bridging may have temporarily restrained upward flow
- **Operational complexity:** Surface losses, heavy cuttings, frequent dump/dilution cycles made it difficult to detect subtle gains or flow indicators through conventional pit volume monitoring

These conditions may have created a deceptive sense of hydraulic stability until the moment of cement placement, when the well encountered its first true pressure exposure.

Well-B

Incident Summary. Following the learnings from Well-A, mitigation strategies were applied in subsequent wells (Well-B & C) to avoid encountering similar incidents and validate solutions for subsequent development well designs. These operations offered new insights and highlighted evolving risks associated with artesian flow. Unlike Well-A, Wells B & C (shown in Fig. 5) are situated in the heart of the Umm as Samim Sabkha.

Despite the mitigations in place, initial challenges in Well-B emerged with simultaneous losses at 190 m (in Rus) and gains at 227 m. Kill brine (11.7 kPa/m) was displaced per the decision tree, and losses ceased, indicating classic loss/gain behavior. However, losses reappeared at 346 m, followed by artesian flow near the Koomey (Accumulator) unit. H₂S levels measured at the cellar peaked to 250 ppm, raising safety concerns. Attempts to displace with kill fluid were hindered by severe losses.

Remedial Actions. The drilling BHA was pulled, and a cement stinger was run in hole. A fast-set cement plug was spotted, temporarily halting the flow. When artesian flow resumed, a second cement plug was deployed, successfully containing the well allowing section drilling to resume without further incidents.

The event resulted in 5.52 days of non-productive time (NPT). Notably, ground subsidence occurred around the well pad due to forklift movement after the leak path dried. A civil team was mobilized for site rectification.

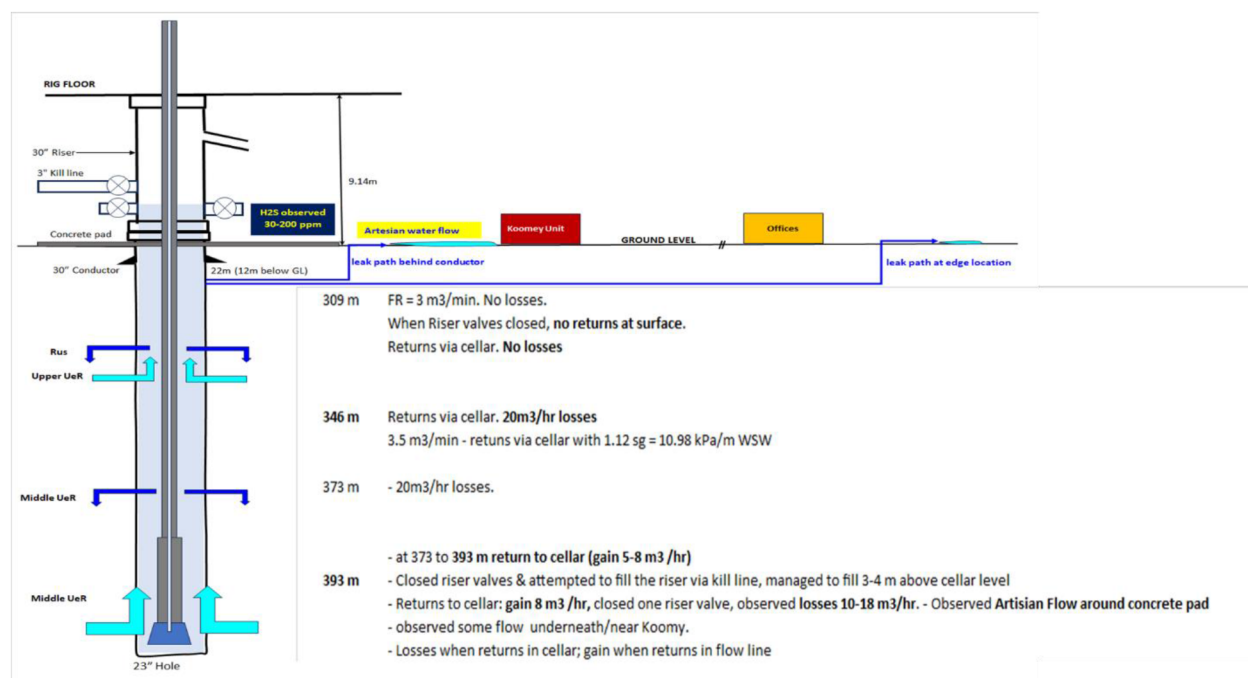


Figure 7—Well schematic with commentary highlights of the operation summary, depths of events, and areas impacted by the incident

Post-Incident Analysis. The Well-B incident marked a shift in artesian flow risk, with flow surfacing near rig-critical areas (e.g., Koomey unit), elevating safety and integrity concerns. H₂S levels reached 250 ppm at the cellar, highlighting the need for gas monitoring and emergency protocols.

Loss/gain behavior and surface subsidence indicated weak formation integrity above the artesian source, reinforcing the need for early reinforcement before zone exposure. Fast-set cement plugs provided temporary control but were limited by placement reliability and introduced NPT.

Post-flow ground movement emphasized the importance of factoring surface stability into engineering plans. Overall, the incident underscored the value of proactive mitigation, such as deeper conductor casing or early surface plugs, to isolate flow-prone zones and reduce reactive intervention.



Figure 8—Artesian flow leak towards koomey area



Figure 9—Subsidence due to forklift movement post artesian flow

Integrated Analysis Approach

Hydrogeology Insights from Literature

While numerous wells were successfully drilled across these aquifers within SDO's operated blocks without issue, the consecutive artesian flow incidents occurred within a single localized area. This prompted a focused investigation to understand the root cause of these incidents. A regional review was conducted, integrating drilling data, incident records, and hydrogeological insights. Particular attention was given to understand the nature of the UeR aquifer by examining available literature and analyzing pore pressure to assess its behavior in the affected area.

Within the regional aquifer system, the Upper and Middle units of the UeR Formation are recognized as the most extensive and hydraulically significant aquifers. In contrast, the Rus Formation often acts as an aquitard. However, permeable zones can exist where dolomitization or carbonate interbeds occur. (United Nations, 2013) Combined with the underlying UER Shammar shale, the aquifer is characterized as a confined aquifer system. The UER aquifer is one of the primary groundwater sources within Oman's hydrogeological basin, believed to be recharged by elevated terrains in the Oman Mountains to the north and the Dhofar Mountains to the south (Al Lamki et al., 1996). The Upper and Middle UeR units serve as the main water-bearing zones and have been extensively utilized in the oil and gas industry as a water source.

Within the SDO's exploration area, very limited data has been recorded and due to limited drilling. Several studies on the hydrogeological aspects of the UeR were reviewed to help explain the behavior of the aquifer regionally. Studies done by [Al Lamki et al. \(1996\)](#) and [Al-Mashaikhi et al. \(2012\)](#) indicate that the UeR aquifers are exposed in the core of the Dhofar mountain to the South and are considered likely to accept direct recharge resulting from the seasonal Khareef rains. Regional aquifer trends such as salinity developed by [Al Lamki et al. \(1996\)](#) suggest that recharge of UeR aquifers from Saudi Arabia through Sahmah. The groundwater is moving south-eastwards and north-eastwards across Oman's Western border from this source area in Saudi Arabia. However, the data suggests that the aquifers within SDO's exploration area, specifically within the Umm as Samim, are assumed to be least impacted by modern day recharge ([Al Lamki et al., 1996](#)). Furthermore, [Al-Mashaikhi et al. \(2012\)](#) suggests that aquifer pressure is expected to increase along the underground waterflow direction from the recharge areas, and there are areas in which the UeR aquifer is fully confined where artesian flow occurs with high pressure.

This information has steered the team toward a deeper investigation of the aquifer's fluid properties and recorded pressures, aiming to reconcile any trends with the insights derived from the study's observations.

Data Integration and Analysis

To first tackle the problem, Heat Maps were employed as a diagnostic tool to visualize trends across different phases of drilling and cementing. By mapping key variables – such as pit volume changes, fluid densities, and pressure offsets – against depth or time, heat maps provide an intuitive overview of how subsurface behavior evolves. The method was used primarily to investigate whether a correlation exists between observed drilling anomalies and specific operational conditions. Key areas explored include:

- **Pressure imbalance triggers:** Highlighting zones where ESD/ECD fell below formation pressure and how close these thresholds were to observed flow events
- **Fluid property transitions:** Identifying patterns in density dropouts that may be related to dilution, artesian influx, or mud contamination ("mud flipping")
- **Event-time relationships:** Examining if pressure shifts coincided with known activities such as riser draining, casing cementation, or pit volume adjustments

Table 2—Heat Map of wells within SDO's exploration area

Well	GL elevation (m)	Spud month	Losses prior UeR Middle?	Artesian Flow and when	MW (kPa/m)
Well-X1	59.64	October	YES, 25 m3/hr but reduced to 3 m3/hr post UeR	YES, while running 18-5/8" CSG	11
Well-X2	72.79	June	NO	YES, while running 18-5/8" CSG	Not Available
Well-X3	59.29	February	YES, 150 m3/hr but reduced to 4 m3/hr post UeR	YES, while drilling UeR observed water flowing near rig camp and sub base. Spotted cement via GRE pipe	11.66
Well-X4	91.44	January	NO	NO	Not Available
Well-X5	59.17	September	YES, 77 m3/hr but reduced to 10 m3/hr post UeR	YES, while drilling in Rus observed flow behind the cellar. Spotted cement via GRE pipe	Not Available
Well-X6	89.68	January	YES, 35 m3/hr but reduced to 6 m3/hr post UeR	NO	11.5
Well-X7	60.02	May	NO	YES, after completing 18-5/8" CSG cement job	11.38

Key Observations:

- Artesian flow mostly occurred below 61 m elevation, except Well-X2
- Rus formation may contribute to artesian flow (e.g. Well-X5)
- 4 wells showed pre-UER losses; losses often reduced post-UER
- Surface flow behind cellar observed in 2 cases
- Mud weights ranged from 11.0–11.66 kPa/m

Collected data from drilling records of wells drilled within and offset of SDO's operated areas was evaluated to identify regional patterns. Key parameters such as estimated pressure gradients (in red), artesian flow events (in yellow), and measured H₂S (in black) were mapped in Fig. 10 below in attempt to identify such trends. The data was plotted over the UeR contour top structure map to correlate data with depth and identify elevated areas of the formation that could be linked with recharging or providing enough pressure head to explain the observed overpressures. As seen from the contour map, minimal elevation change is observed in the UeR formation indicating no near outcrops ruling out the possibility of recharging during the rainy season. A key observation from data trends indicates artesian flow occurrences mostly occurred from well penetrations within the Area-3 & 4. It could also be inferred from the estimated pressure gradients Areas 3&4 exhibit much higher pore pressures than in Area-1.

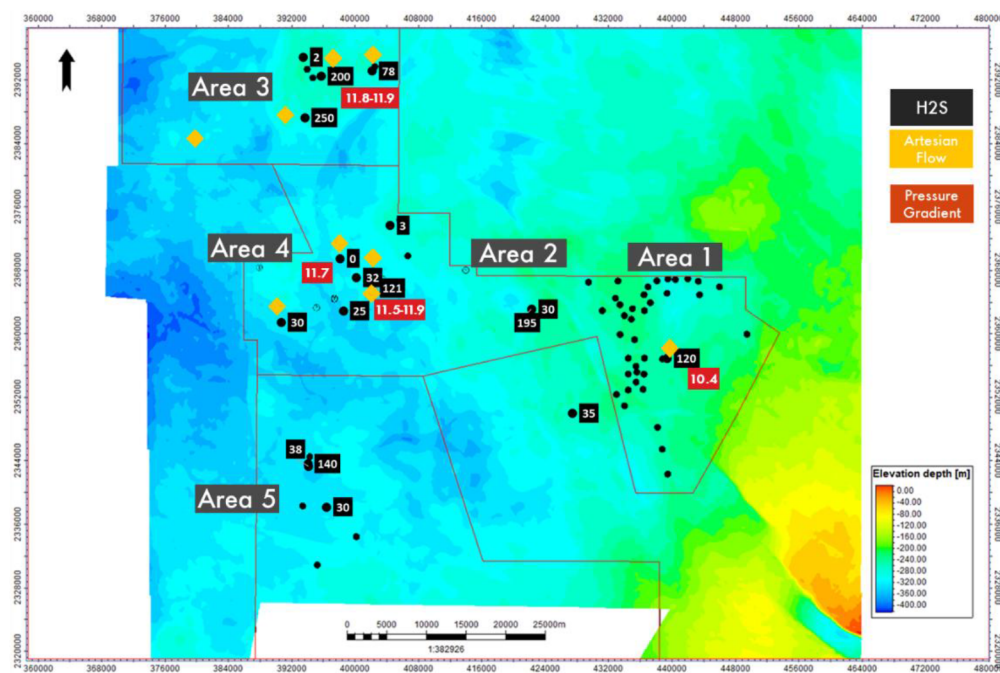


Figure 10—Top structure map of the UeR formation highlighting estimated pressure gradients (in red), recorded artesian flow (in yellow), and measured H₂S (in black) from wells drilled within SDO's operated areas

An in-depth analysis of the pore pressures was done to investigate the aquifer pressure system across the different areas. Based on the Fig. 11 below, it can be clearly interpreted that wells drilled in Areas 3&4 follow the same pressure gradient and likely sharing a hydraulic system that is different from Area 1 and at much higher pressure.

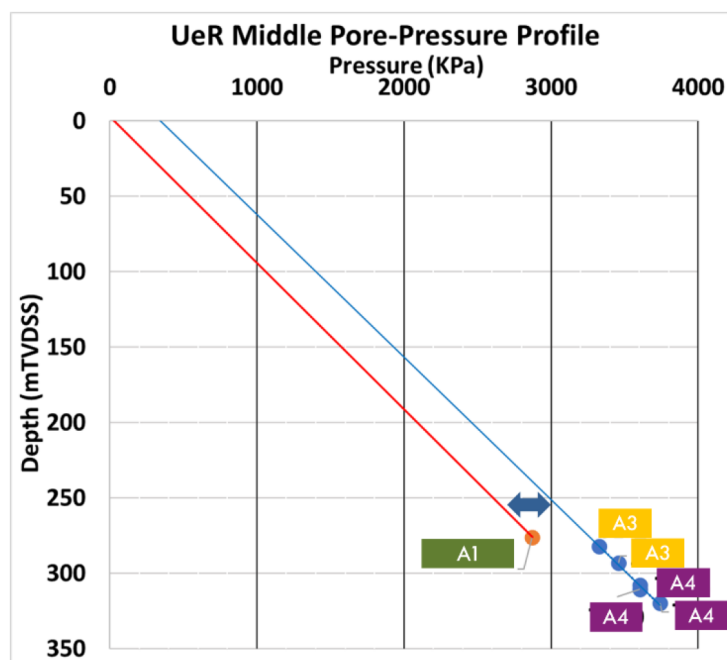


Figure 11—Estimated pore pressure evaluation from wells drilled across different areas indicating different pressure systems between Area 1 and Areas 3&4

Further evidence from fluid analysis done on water samples collected from wells with artesian flow incidents indicated the likelihood of two distinct aquifer systems across SDO's operated areas. This can be seen from water density and salinity data in Table 3 below highlighting the high contrast in aquifer water properties between Area 1 and Areas 3&4 which are notably within similar depth.

Table 3—Fluid property data from artesian flow water samples

Water Property	Area-1	Area-3&4
Density [g/cc]	1.055	1.085
Salinity [ppm]	80k-90k	120k -170k

The significant contrast in water salinity and density between Area 1 and Areas 3&4 suggests the presence of hydraulically distinct aquifer systems. The above findings have direct implications on well planning in Areas 3&4 due to elevated pore pressures and increased artesian flow risk. Future well planning must incorporate these localized fluid properties to ensure accurate pressure predictions and safe drilling operations.

A water table analysis (Fig. 12) was used to create pore pressure scenarios assuming different aquifer tops integrating the identified pressure gradient and water property ranges. The analysis provides an estimate of the water table depth above mean-sea level that would reflect data described above which will then be used to calculate the pressure prediction ranges. Based on the chart below, in almost all cases, the Wells 1 to 5 drilled within Areas 3&4 indicate the aquifer would deliver a pressure head of 95-140 m compared to 60-100m in Well-6 drilled in Area-1.

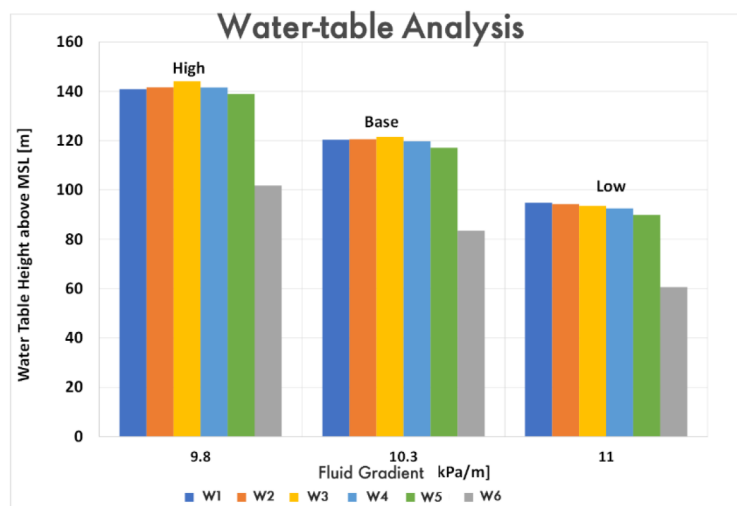


Figure 12—Water Table Analysis to develop pore pressure scenarios using aquifer depth and pressure head ranges

Using the data above, an updated pore pressure prediction is developed honoring the range of possible scenarios for wells drilled across the shallow section within SDO's operated areas. The pore pressure prediction model was developed using a 5-point scenario system ranging from absolute minimum to absolute maximum. This approach integrates aquifer depth, water table elevation, fluid properties, and recorded pore pressures to generate a more robust prediction envelope. Compared to traditional models, this method accounts for wider pressure scenarios including localized aquifer behaviors. Fig. 13 highlights an example of the prediction expressed in Equivalent Mud Gradient (EMG) plotted against depth.

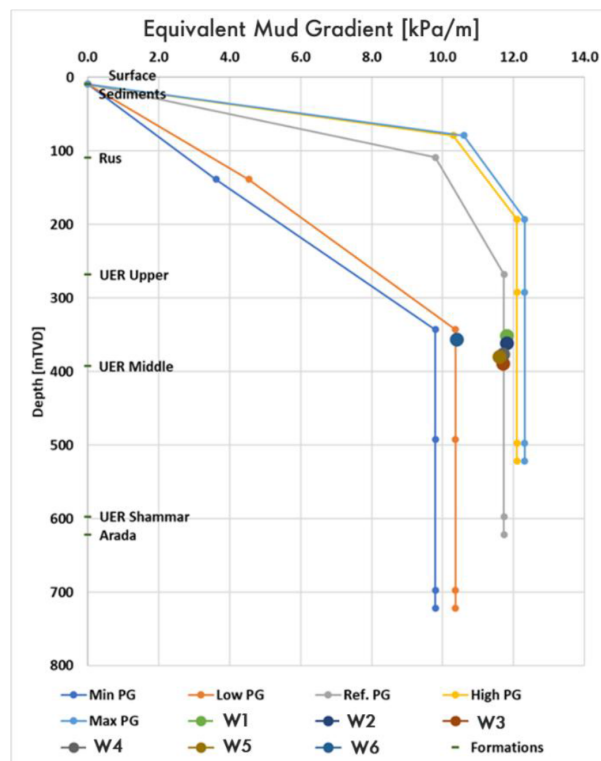


Figure 13—Example of the updated pore pressure prediction range scenarios for the shallow drill section to be used for future wells drilled within SDO's operated areas. The plot shows pressure scenarios using a 5-point system and pressures estimated from the previous wells

Cementing Program Assessment

Cement was designed to surface with the following fluid gradients:

- Spacer: 10 kPa/m
- Lead: 13.8 kPa/m
- Tail: 18.6 kPa/m

Despite lead/tail being overbalanced, the 10 kPa/m spacer, when reaching UER Middle, rendered the well briefly underbalanced, likely triggering the artesian breakthrough as highlighted in Fig. 14. This pressure exposure during float checks aligns with onset of flow. It is worth noting that standard spacer practices across wells had not previously resulted in artesian flow, making this event an anomaly — highlighting the need for a revised pore pressure prediction.

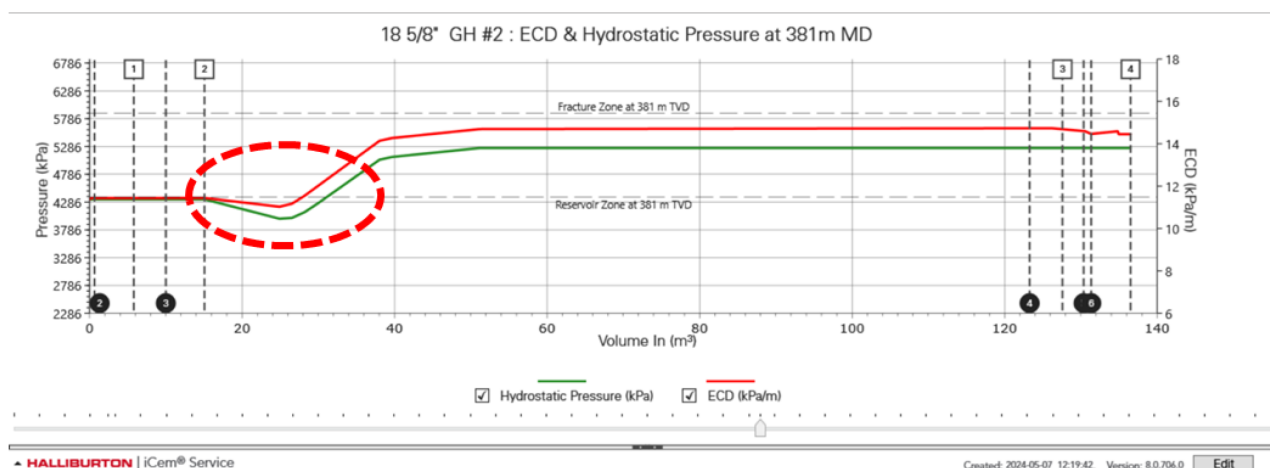


Figure 14—ECD & Hydrostatic Pressure at Middle UER with 10 kPa/m Spacer Density

Mitigation Implementation and Results

The data integration allowed for a well-rounded understanding of the behaviors of the shallow section aquifers. It provided key insights on handling losses and overpressures within the first drill section which was not common within a well-known aquifer within the industry. This is even compounded with drilling in a complex environment such as the Umm as Samim Sabkha of which very few operators have drilled in. This shift from well-level troubleshooting to area-wide analysis enabled a clearer understanding of localized risk and informed targeted mitigation strategies for future wells.

The updated pore pressure prediction provided the grounding for a more robust fluid design. This allowed preparing different scenarios during each phase of the drilling. To control the losses when drilling through the overpressured units, the team proposed to cement the first 100 m of the well section. Following this the cement plug is drilled off under controlled conditions. Furthermore, when preparing to cement the casing, the most critical takeaway from assessment was the need to optimize spacer density. Cementing simulation results in Fig. 15 indicate that a minimum of 13 kPa/m spacer density is required to maintain an overbalanced condition throughout the cementing operation.

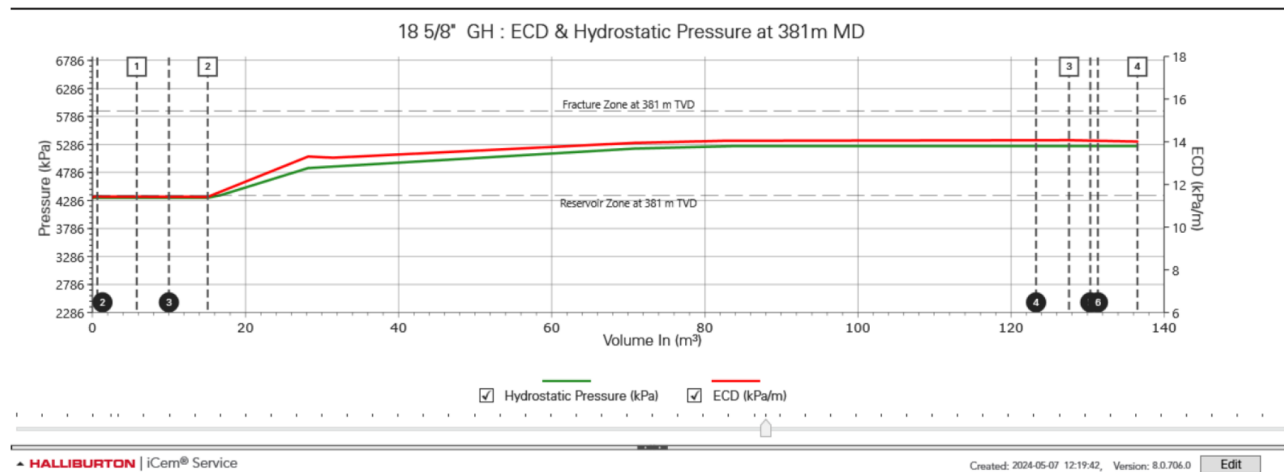


Figure 15—ECD & Hydrostatic Pressure at Middle UER with 13 kPa/m Spacer Density

In addition, the following proactive contingency measures are recommended to mitigate the risk of uncontrolled artesian flow and ensure operational resilience:

- Mobilize 200 m of PVC pipes for immediate deployment
- Modify the mule shoe configuration prior to spud to improve running the spaghetti pipes
- Prepare 400 m of 1.66" spaghetti pipes for contingency circulation pathways
- Pre-mix at least 70 m³ of 11.7 kPa/m kill brine before spud
- Mix 50 m³ of 19.4 kPa/m kill mud for emergency use in case of uncontrolled artesian flow
- Account for riser margin loss by increasing drilling fluid density by 0.27 kPa/m over section total depth

These actions strengthen hydraulic control strategies during critical phases of surface drilling and cementing.

Based on the mitigations above Well-C, drilled within the heart of the Sabkha, was drilled safely and efficiently without any artesian flow observed across the shallow drill section, reducing the NPT from 7.5 days to 2.5 days (~67% reduction) compared to Well-A as highlighted in Fig. 16 below.

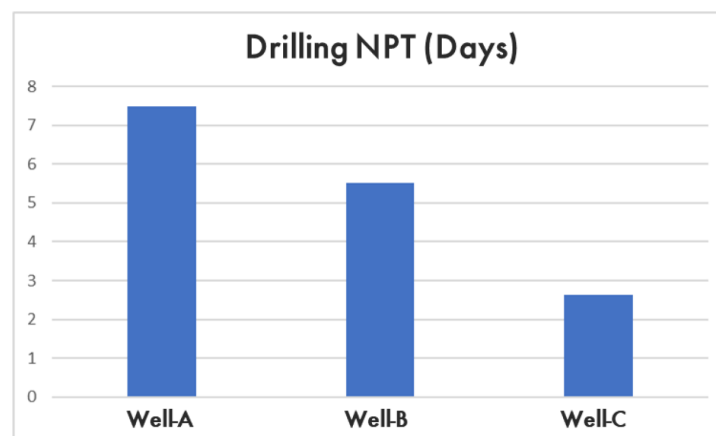


Figure 16—Progression in diminishing Non-Productive Time (NPT) after implementing the learnings from Well-A&B in Well-C leading to a drop from ~7.5 to 2.5 days

Furthermore, the team has looked into additional long-term mitigations to further reduce NPT related to remedial measures especially for wells drilled in Sabkha. As a learning from the incident in Well-B, it is critical to strengthen the very first portion of the hole even before drilling through the aquifer to reduce reliance on remedial cement plugs. A proposed solution is to deepen the conductor casing which in turn would lead to enhancing wellbore integrity, improving safety margins, and reducing civil remediation time and cost

Conclusion

This case study from Shell Development Oman's operations in the Umm as Samim Sabkha highlights the critical importance of integrating subsurface, and operational data to overcome the unique challenges of drilling through shallow, overpressured, and sour aquifers in sabkha environments. The incidents encountered in Wells A and B underscore the risks posed by underbalanced conditions, artesian flow, and H₂S exposure, particularly in areas with limited subsurface data and complex hydrogeological behavior.

Through a comprehensive post-incident analysis, including pressure back-calculations, heat map diagnostics, and fluid property evaluations, the team identified key triggers and developed a revised pore pressure prediction model tailored to localized aquifer systems. This enabled the implementation of targeted mitigation strategies, such as optimized spacer densities, proactive well control preparations, and contingency equipment deployment.

The successful execution of Well-C, with no artesian flow and a 67% reduction in non-productive time compared to Well-A, validates the effectiveness of this integrated, data-driven approach. Furthermore, the study emphasizes the value of transitioning from reactive troubleshooting to predictive planning, offering a replicable framework for future drilling campaigns in similarly challenging environments.

Abbreviations

BHA:	Bottom Hole Assembly
CMT:	Cement
ECD:	Equivalent Circulating Density
EMG:	Equivalent Mud Gradient
ESD:	Equivalent Static Density
GL:	Ground Level
GRE:	Glass Reinforced Epoxy
H ₂ S:	Hydrogen Sulfide
MSL:	Mean Sea Level
MW:	Mud Weight
NPT:	Non-Productive Time
POOH:	Pull Out of Hole
PP:	Pore Pressure
PVC:	Polyvinyl Chloride
RIH:	Run In Hole
SDO:	Shell Development Oman
SRB:	Sulfate Reducing Bacteria
UeR:	Umm er Radhuma

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